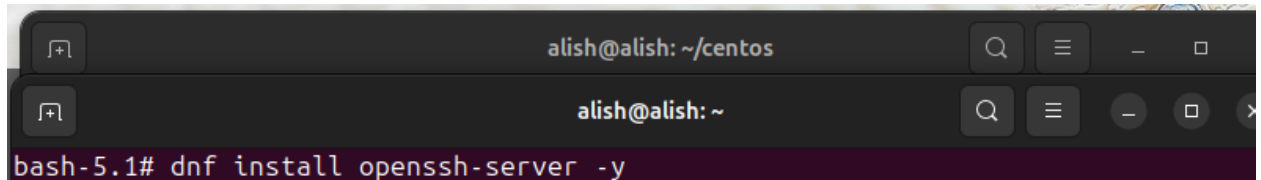


Title: Configure SSH Server to allow/deny root login and allow/deny user login

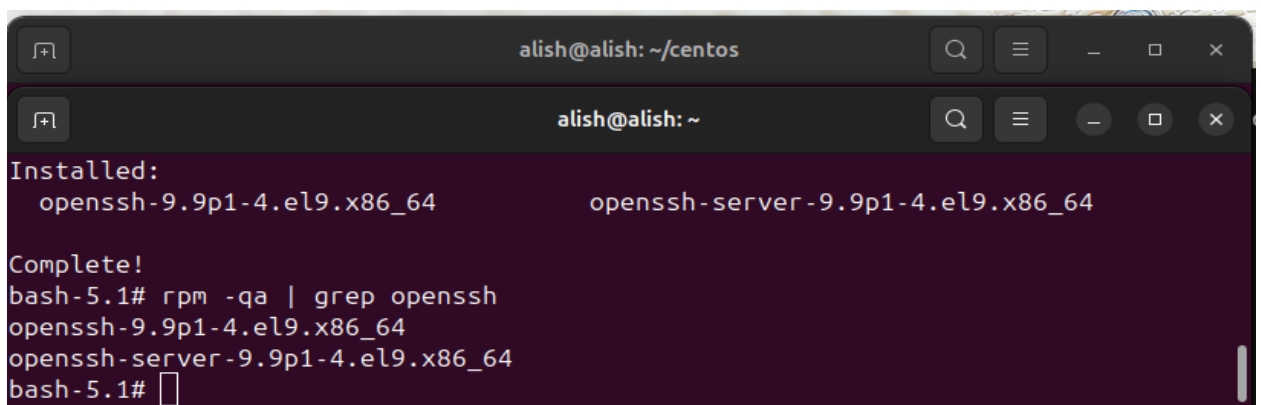
A. Install required package for OpenSSH server

Command : – dnf install openssh-server –y

– rpm –qa | grep openssh



```
alish@alish: ~/centos
alish@alish: ~
bash-5.1# dnf install openssh-server -y
```



```
alish@alish: ~/centos
alish@alish: ~
Installed:
  openssh-9.9p1-4.el9.x86_64      openssh-server-9.9p1-4.el9.x86_64

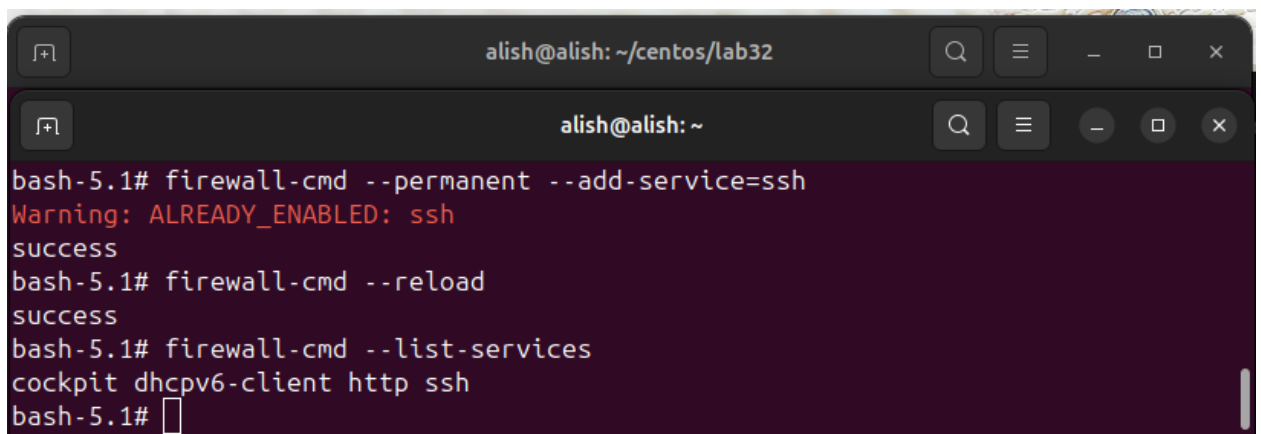
Complete!
bash-5.1# rpm -qa | grep openssh
openssh-9.9p1-4.el9.x86_64
openssh-server-9.9p1-4.el9.x86_64
bash-5.1#
```

B. Allow such packets to enter through the firewall

Command : – firewall-cmd – permanent – add-service=ssh

– firewall-cmd – reload

– firewall-cmd – list-services



```
alish@alish: ~/centos/lab32
alish@alish: ~
bash-5.1# firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
bash-5.1# firewall-cmd --reload
success
bash-5.1# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
bash-5.1#
```

C. Start and enable ssh service.

Command- ssh-keygen -A

- /usr/sbin/sshd

- ps aux | grep sshd

```
alish@alish: ~/centos/lab32
alish@alish: ~
bash-5.1# ssh-keygen -A
ssh-keygen: generating new host keys: RSA ECDSA ED25519
bash-5.1# /usr/sbin/sshd
```

```
alish@alish: ~/centos/lab32
alish@alish: ~
bash-5.1# ps aux | grep sshd
root      460  0.0  0.0  11780  3260 ?        Ss   03:06   0:00 sshd: /usr/s
in/sshd [listener] 0 of 10-100 startups
root      470  0.0  0.0   3348  1772 pts/0    S+   03:07   0:00 grep sshd
bash-5.1# ps aux | grep sshd
root      460  0.0  0.0  11780  3260 ?        Ss   03:06   0:00 sshd: /usr/s
in/sshd [listener] 0 of 10-100 startups
root      472  0.0  0.0   3348  1764 pts/0    S+   03:07   0:00 grep sshd
bash-5.1#
```

D. Configure SSH server to deny direct root login

Command – nano /etc/ssh/sshd_config

– Once inside sshd_config do ctrl+W and search for PermitRootlogin we will likely see a line that says #PermitRootLogin prohibit-password or PermitRootLogin yes.

- We will change it to PermitRootLogin no
- Save it and exit (ctrl+O enter and ctrl+X)

```
bash-5.1# nano /etc/ssh/sshd_config
```

```
alish@alish: ~/centos/lab32
alish@alish: ~
GNU nano 5.6.1 /etc/ssh/sshd_config
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

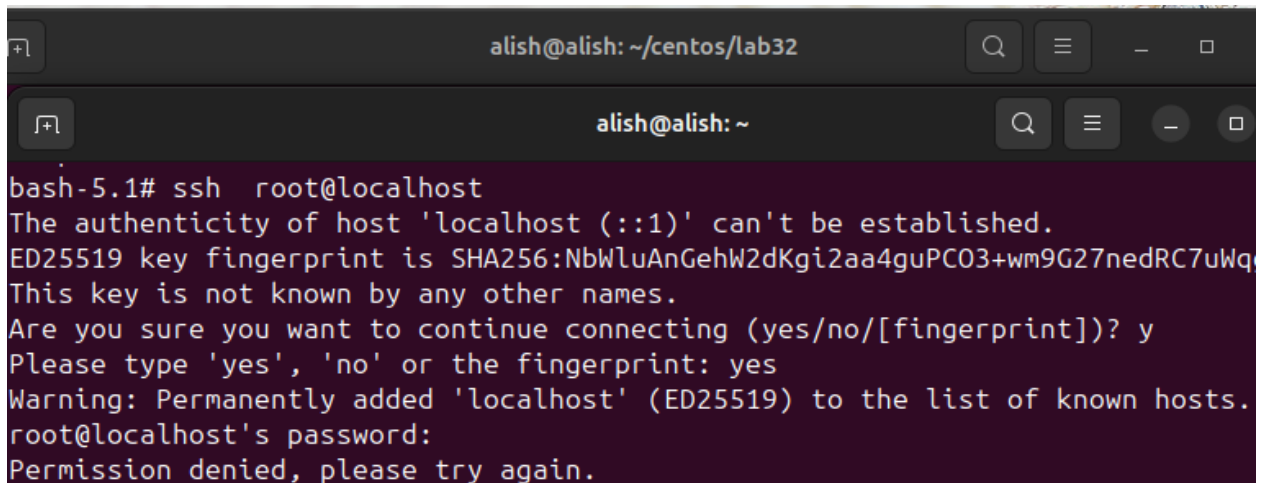
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

- Pkill sshd && /usr/sbin/sshd

- ssh root@localhost

Even the root can't login in, as we have deny access to root user too even with a correct passwd



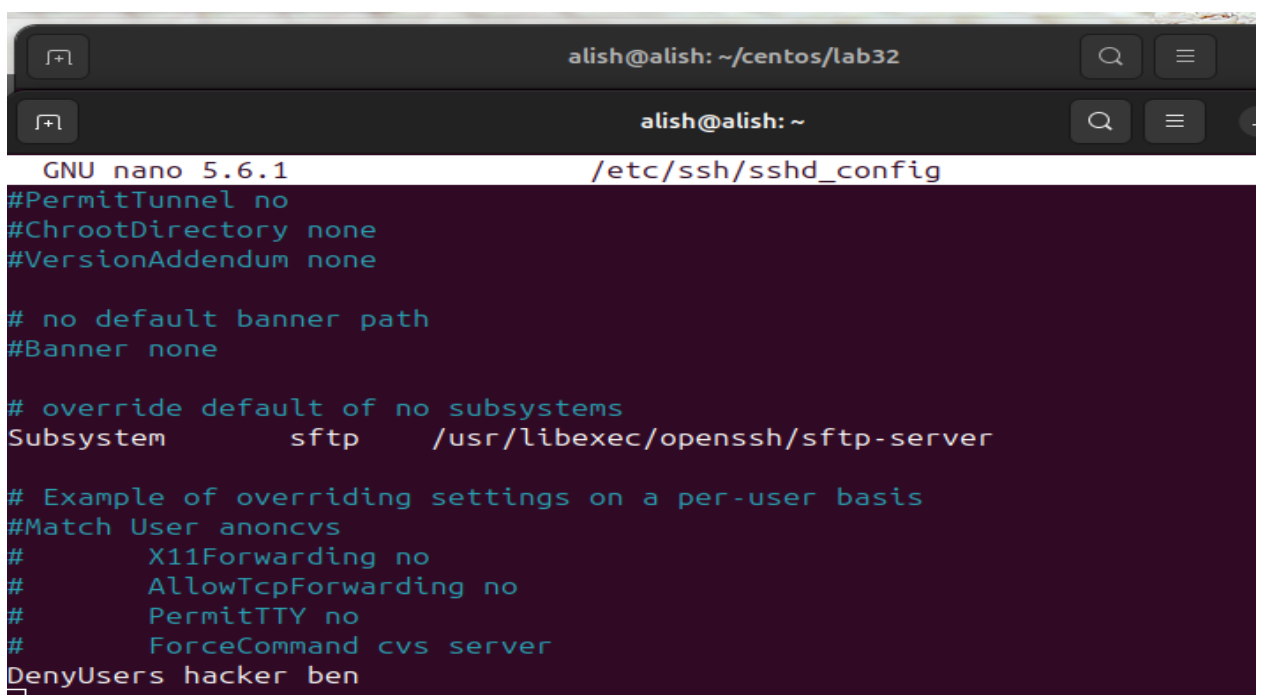
```
alish@alish: ~/centos/lab32
alish@alish: ~
bash-5.1# ssh root@localhost
The authenticity of host 'localhost (:::1)' can't be established.
ED25519 key fingerprint is SHA256:NbWluAnGehW2dKgi2aa4guPC03+wm9G27nedRC7uWq
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
root@localhost's password:
Permission denied, please try again.
```

E. Configure open SSH server to block login from user

Command : – nano /etc/sbin/sshd_config

- once you are inside the sshd_config go to the last line and add DenyPermit hacker
- pkill sshd
- /usr/sbin/sshd
- useradd hacker && passwd hacker

```
bash-5.1# nano /etc/ssh/sshd_config
```



```
alish@alish: ~/centos/lab32
alish@alish: ~
GNU nano 5.6.1 /etc/ssh/sshd_config
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

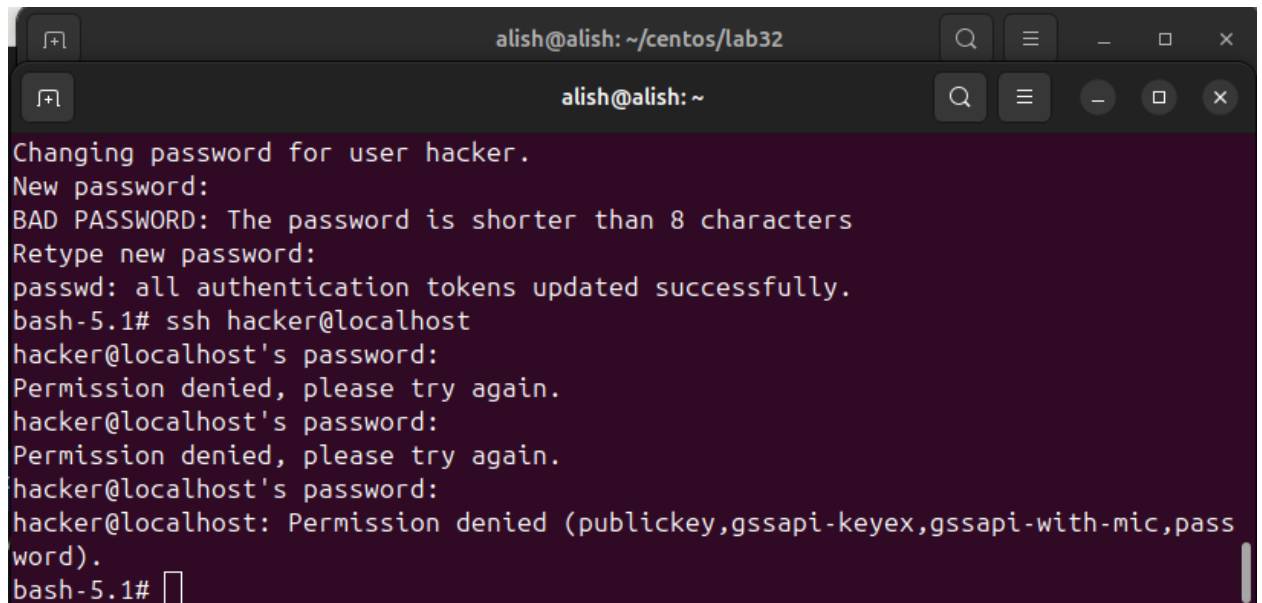
# no default banner path
#Banner none

# override default of no subsystems
Subsystem sftp /usr/libexec/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
DenyUsers hacker ben
```

- ssh hacker@localhost

Even if hacker type correct password we have deny hacker user

A terminal window with a dark background and light-colored text. The window title is 'alish@alish: ~/centos/lab32'. The terminal shows the following sequence of commands and outputs: 1. 'Changing password for user hacker.' 2. 'New password:' followed by 'BAD PASSWORD: The password is shorter than 8 characters'. 3. 'Retype new password:' followed by 'passwd: all authentication tokens updated successfully.'. 4. 'bash-5.1# ssh hacker@localhost'. 5. 'hacker@localhost's password:' followed by 'Permission denied, please try again.'. 6. 'hacker@localhost's password:' followed by 'Permission denied, please try again.'. 7. 'hacker@localhost's password:' followed by 'hacker@localhost: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password)'. 8. The prompt 'bash-5.1#' is shown at the end with a cursor. The terminal window has standard Linux window controls (minimize, maximize, close) and a search icon in the top right corner.

```
alish@alish: ~/centos/lab32
alish@alish: ~
Changing password for user hacker.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
bash-5.1# ssh hacker@localhost
hacker@localhost's password:
Permission denied, please try again.
hacker@localhost's password:
Permission denied, please try again.
hacker@localhost's password:
hacker@localhost: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
bash-5.1#
```